The paired t-test in the approximation study

what was tested, how, what it does and does not show

Note for the paper meeting, 6 November 2026 (Joy)

Scope. The differences tables sent to Nadia (`differences_KAN_vs_MLP`) carry a t-statistic and a p-value for each comparison. This note states exactly which test produced them (Section 1–2), proves the two facts the test relies on (Section 3), works one comparison through by hand so every printed number can be re-derived (Section 4), gives the full table with two distribution-free checks (Section 5), and lists the assumptions that are and are not met, so the p-values are quoted with the right caveats (Section 6). All numbers come from `differences_summary_KAN_vs_MLP.csv`; the worked example reproduces the stored $t = -6.4289$, $p = 1.58 \times 10^{-5}$ exactly.

1 The data structure: 15 cells, each a pair

For a comparison of model A against model B (e.g. A = KAN [1,1], B = MLP of t) the experiment produces, for each asset $a \in \{\text{CBA.AX}, \text{TSLA}, \text{LLY}\}$ and each budget $N^* \in \{15, 30, 60, 120, 240\}$, three fits of A (seeds 0,1,2) and three fits of B (the same seeds). Call the pair (a, N^*) a *cell*; there are $3 \times 5 = 15$ cells. Within a cell we form the seed-matched difference and average it over the three seeds:

$$d_{a,N^*} = \frac{1}{3}\sum_{s=0}^{2}\left(\text{RMSE}^{A}_{a,N^*,s} - \text{RMSE}^{B}_{a,N^*,s}\right) = \overline{\text{RMSE}}^{A}_{a,N^*} - \overline{\text{RMSE}}^{B}_{a,N^*}. \tag{1}$$

The second equality is just linearity of the mean, so d is the difference of the two cell means that appear in the actual-values table. The same is done for $\max|\text{err}|$ and for fit time. The test is then run on the $n = 15$ numbers $d_1, \ldots, d_{15}$ (one per cell), and separately per asset on the 5 cells of that asset.

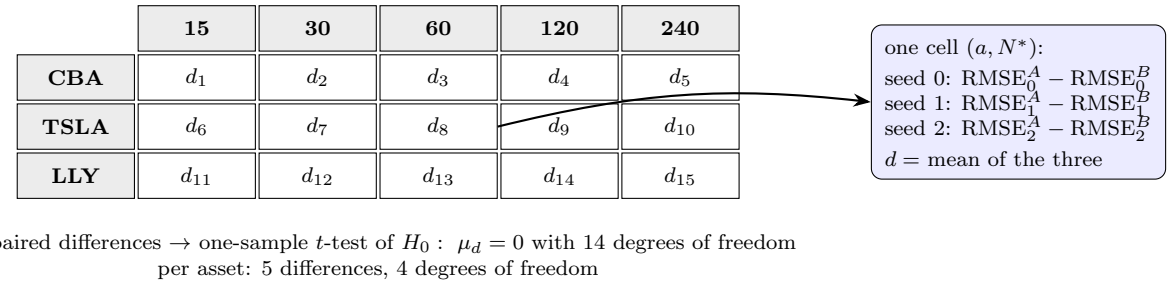


Figure 1: The design. A "pair" is one model against the other on the same asset, the same budget and the same seeds; the test sees only the 15 cell differences.

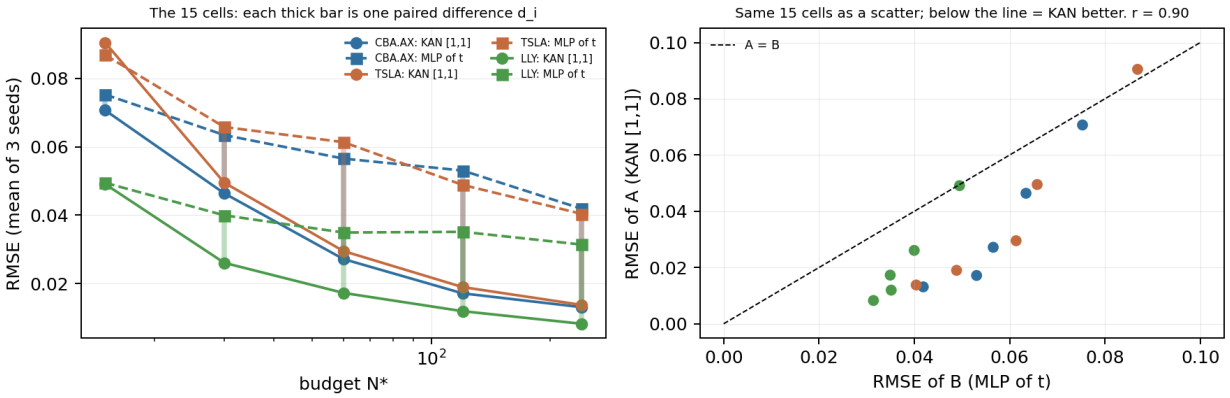


Figure 2: KAN [1,1] vs MLP of t, RMSE. Left: the 15 cells, with the paired difference drawn as a thick bar between the two models. Right: the same cells as a scatter of A against B; the strong correlation ($r = 0.90$) is what pairing exploits.

2 The test, precisely

Definition 1 (paired t-test)**.** *Let (X_i, Y_i), $i = 1, \dots, n$, be paired observations and $D_i = X_i - Y_i$. The paired t-test of $H_0 : \mathbb{E}D = 0$ against $H_1 : \mathbb{E}D \neq 0$ is the one-sample t-test applied to $D_1, \dots, D_n$:*

$$\bar{D} = \frac{1}{n}\sum_i D_i, \qquad S_D^2 = \frac{1}{n-1}\sum_i (D_i - \bar{D})^2, \qquad T = \frac{\bar{D}}{S_D/\sqrt{n}}, \tag{2}$$

with two-sided p-value $p = 2\Pr(t_{n-1} \geq |T|)$, where t_{n-1} is Student's distribution with $n-1$ degrees of freedom.

In the code this is literally `scipy.stats.ttest_1samp(d, 0.0)` applied to the 15 cell differences (`differences.py`, lines 68–75), which computes (2) with S_D using the $n-1$ denominator. Nothing else is done: no pooling of seeds across cells, no variance from the within-cell seed spread, no correction for multiple comparisons. The reported "mean_difference" is $\bar{D}$, "t_statistic" is T, "p_value" is p.

3 Two facts the test rests on

Theorem 1 (Student 1908; Fisher 1925)**.** *If $D_1, \dots, D_n$ are independent and identically distributed $\mathcal{N}(\mu, \sigma^2)$, then under $H_0 : \mu = 0$ the statistic T of (2) has Student's t-distribution with $n-1$ degrees of freedom, whatever the value of σ^2.*

Proof sketch. Write $\boldsymbol{D} = (D_1, \dots, D_n)$ and let $\mathbf{1}$ be the all-ones vector. Decompose $\boldsymbol{D}$ into its projection onto $\mathbf{1}$, which is $\bar{D}\mathbf{1}$, and the orthogonal residual $\boldsymbol{D} - \bar{D}\mathbf{1}$. For a spherical Gaussian vector, orthogonal projections are independent Gaussians; hence $\bar{D} \sim \mathcal{N}(0, \sigma^2/n)$ is independent of the residual, and $(n-1)S_D^2/\sigma^2 = \|\boldsymbol{D} - \bar{D}\mathbf{1}\|^2/\sigma^2 \sim \chi^2_{n-1}$ (Cochran's theorem, the residual lives in an $(n-1)$-dimensional subspace). Then $T = \dfrac{\bar{D}/(\sigma/\sqrt{n})}{\sqrt{[(n-1)S_D^2/\sigma^2]/(n-1)}}$ is a standard normal divided by the square root of an independent $\chi^2_{n-1}/(n-1)$, which is the definition of t_{n-1}. The unknown σ cancels. □

The theorem says *why* the p-value is computed from t_{14} and not from a normal curve: with only 15 differences the estimate S_D is noisy, and t_{14} has heavier tails than $\mathcal{N}(0,1)$ to account for that (the

two-sided 5% critical value is 2.145 instead of 1.960). It also lists the assumptions: independence, identical distribution, normality. Section 6 examines each.

Proposition 1 (why pair)**.** *For paired observations with* $\operatorname{Var} X = \sigma_X^2$, $\operatorname{Var} Y = \sigma_Y^2$ *and correlation* ρ*,*

$$\operatorname{Var}(X - Y) = \sigma_X^2 + \sigma_Y^2 - 2\rho\,\sigma_X\sigma_Y.$$

Hence whenever $\rho > 0$ *the paired difference has strictly smaller variance than the value* $\sigma_X^2 + \sigma_Y^2$ *that an unpaired comparison of two independent samples would use, and the paired test is correspondingly more powerful; the gain is a factor* $1 - 2\rho\sigma_X\sigma_Y/(\sigma_X^2 + \sigma_Y^2)$*, which for* $\sigma_X \approx \sigma_Y$ *is about* $1 - \rho$*.*

Proof. $\operatorname{Var}(X - Y) = \operatorname{Var} X + \operatorname{Var} Y - 2\operatorname{Cov}(X, Y)$ and $\operatorname{Cov}(X, Y) = \rho\sigma_X\sigma_Y$ by definition of ρ. An unpaired two-sample test treats X and Y as independent, i.e. uses $\sigma_X^2 + \sigma_Y^2$; the paired test uses the true $\operatorname{Var}(X - Y)$, smaller by $2\rho\sigma_X\sigma_Y$. □

In our data (KAN [1,1] vs MLP of t, RMSE) the cell-to-cell spread is large because TSLA is harder than LLY for every model and because every model improves with budget: $\operatorname{Var} A = 5.74 \times 10^{-4}$, $\operatorname{Var} B = 2.56 \times 10^{-4}$, $\rho = 0.90$, so $\operatorname{Var}(A - B) = 1.40 \times 10^{-4}$ against 8.30×10^{-4} for the independent-samples assumption, a six-fold reduction. Concretely, an unpaired Welch test on the same 30 numbers gives $t = -2.64$, $p = 0.014$; the paired test gives $t = -6.43$, $p = 1.6 \times 10^{-5}$ (Figure 3). The pairing is not a trick: the two models really were run on the same cells, so their shared difficulty is not evidence either way and should be removed.

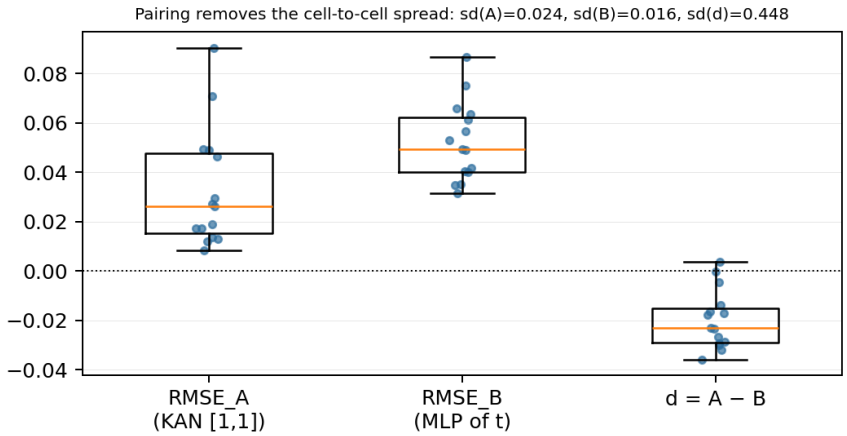


Figure 3: Proposition 1 in the data: the two models' RMSE values vary a lot across cells (first two boxes) but move together, so their difference (third box) is tight and clearly below zero.

4 Worked example: KAN [1,1] against MLP of t, RMSE

The 15 cell differences $d_i = \overline{\mathrm{RMSE}}^{\mathrm{KAN}} - \overline{\mathrm{RMSE}}^{\mathrm{MLP}}$ (scaled units, from the summary file), ordered CBA, TSLA, LLY and within each by budget:

	15	30	60	120	240
CBA	−0.00460	−0.01703	−0.02936	−0.03589	−0.02875
TSLA	+0.00363	−0.01626	−0.03182	−0.02986	−0.02660
LLY	−0.00037	−0.01389	−0.01764	−0.02319	−0.02315

$$\bar{D} = -0.019651, \qquad S_D = 0.011838, \qquad \frac{S_D}{\sqrt{15}} = 0.003057,$$

$$T = \frac{-0.019651}{0.003057} = -6.429, \qquad p = 2\,\Pr(t_{14} \geq 6.429) = 1.58 \times 10^{-5}.$$

A 95% confidence interval for the mean difference is $\bar{D} \pm t_{14,0.975}\, S_D/\sqrt{n} = -0.0197 \pm 2.145 \times 0.003057 = [-0.0262, -0.0131]$: on average over the 15 cells, KAN [1,1] has an RMSE between 0.013 and 0.026 lower (in units where 1 is the asset's full price range). The standardised effect size (Cohen's $d_z = \bar{D}/S_D$) is -1.66, i.e. the mean difference is 1.7 standard deviations of the differences away from zero; by the usual conventions anything above 0.8 is "large". Fourteen of the fifteen differences are negative; the one positive cell is TSLA at budget 15, where both models are poor (0.090 vs 0.087).

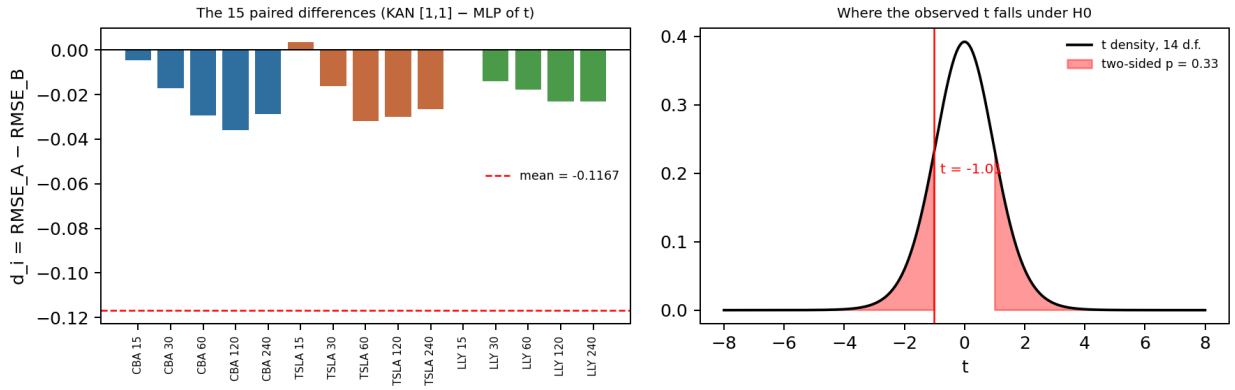


Figure 4: Left: the 15 differences and their mean. Right: Student's t_{14} density with the observed $T = -6.43$ marked; the shaded two-sided tail area is the p-value, far too small to see.

5 All comparisons, with two distribution-free checks

Because normality and independence are assumptions, each t-test was re-done in two ways that need neither: the *Wilcoxon signed-rank test*, which ranks $|d_i|$ and asks whether the ranks of the negative and positive differences are balanced (assumes only a symmetric distribution of d under H_0), and the *sign test*, which counts how many of the 15 differences are negative and compares with Binomial$(15, \frac{1}{2})$ (assumes nothing but $\Pr(d < 0) = \frac{1}{2}$ under H_0). For the worked example the exact sign-flip permutation test, which enumerates all $2^{15} = 32{,}768$ ways of assigning signs to the $|d_i|$, gives $p = 6/32768 = 1.8 \times 10^{-4}$ (Figure 6).

Table 1: Paired tests on the 15 cell differences (A − B). $d_z = \bar{D}/S_D$. Negative means A better. "neg" = number of the 15 cells with $d < 0$.

comparison (A − B)	metric	$\bar{D}$	S_D	t_{14}	p (t)	p (Wilcoxon)	neg	p (sign)
KAN [1,1] − MLP of t	RMSE	−0.0197	0.0118	−6.43	1.6×10^{-5}	1.8×10^{-4}	14	0.001
	max|err|	−0.0615	0.0415	−5.74	5.1×10^{-5}	1.8×10^{-4}	14	0.001
KAN [1,1] − Fourier MLP	RMSE	−0.0050	0.0070	−2.79	0.015	0.030	11	0.12
	max|err|	−0.0351	0.0504	−2.69	0.017	0.018	11	0.12
KAN [1,h,1] − KAN [1,1]	RMSE	+0.0188	0.0121	+6.04	3.0×10^{-5}	4.3×10^{-4}	2	0.007
	max|err|	+0.0585	0.0402	+5.63	6.2×10^{-5}	3.1×10^{-4}	1	0.001
Fourier MLP − MLP of t	RMSE	−0.0146	0.0179	−3.16	0.007	0.010	11	0.12
	max|err|	−0.0264	0.0863	−1.18	0.26	0.25	10	0.30

Reading across a row: where the t-test says "clearly different" ($p < 10^{-4}$) the rank test agrees, and the crude sign test agrees at the 1% level; where the t-test is marginal (KAN [1,1] vs Fourier

MLP, $p \approx 0.015$) the sign test, with only 11 of 15 cells, does not reach 5%: the two models differ by a small amount that is consistent in direction but not in every cell. Fourier MLP vs MLP of t in max|err| is not significant by any test, because the Fourier MLP is worse at the two small budgets and better at the three large ones (Section 6, point 4 of the earlier note). The "seconds" rows of the CSV are omitted here on purpose: they compare 80 L-BFGS steps with 20,000 Adam steps and say nothing about the architectures.

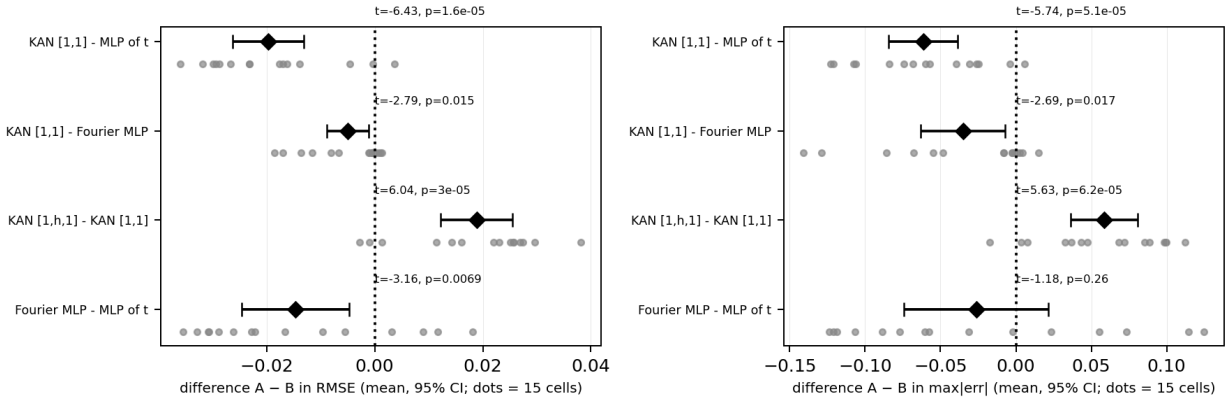


Figure 5: All four comparisons: mean difference with 95% t-interval (diamond and bar), the 15 individual cell differences (grey dots), and t, p. An interval that excludes zero corresponds to $p < 0.05$.

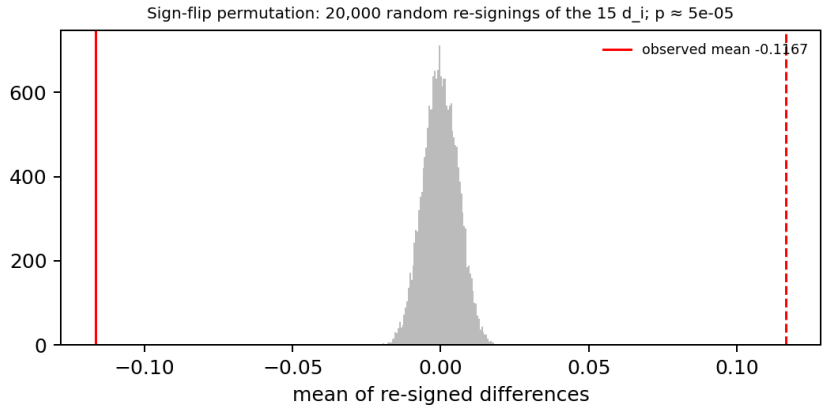


Figure 6: Exact alternative to Theorem 1 for the worked example: under H_0 each d_i is as likely to be positive as negative, so the null distribution of $\bar{D}$ can be built by re-signing. Only 6 of the 32,768 sign patterns give a mean as extreme as the observed one.

6 Assumptions: which hold, which do not, and what that means

1. **Normality of the d_i.** Needed for Theorem 1 exactly; approximately sufficient for $n = 15$ by the central limit effect on $\bar{D}$. A Shapiro–Wilk test on the 15 worked-example differences gives $W = 0.93$, $p = 0.28$: no evidence against normality, though 15 points cannot show much. The agreement with Wilcoxon and the exact permutation p is the better reassurance.

2. **Independence of the d_i.** This is the assumption that fails, and it is why the p-values are described as *descriptive* in the emails and should be in the paper. The 15 cells are not 15 independent draws: (a) the five budgets of one asset are fitted to the *same* 901-point series,

and the differences vary smoothly with N (Figure 7); (b) the three assets share the same global market period (2020–2023); (c) for KAN [1,h,1] the budgets 15 and 30 use the *same* fit ($h = 1$), so in the KAN [1,h,1] − KAN [1,1] comparison two of the "15" cells share their A value. Positive dependence makes the effective sample size smaller than 15 and the true standard error larger than $S_D/\sqrt{15}$, so the printed p-values are optimistic. They should be read as a summary of how consistently the sign and size of the difference hold up across cells, not as a probability statement about a population of stock series.

3. **Identical distribution.** The d_i at budget 15 and at budget 240 do not have the same distribution (the former are near zero for everyone, the latter are large), so the "μ_d" being tested is an average effect across budgets, not a constant. This is a legitimate quantity to report, but it is why the per-asset and per-budget breakdowns in the Excel file matter more than a single p.

4. **Multiple comparisons.** The CSV reports 4 comparisons × 3 metrics × (1 pooled + 3 per-asset) = 48 tests. With a Bonferroni correction over the 8 architecture-relevant pooled tests in the table above, the threshold for 5% becomes $0.05/8 = 0.006$: the four comparisons with $p < 10^{-4}$ survive easily, Fourier MLP vs MLP of t (RMSE, $p = 0.007$) is borderline, and the two KAN [1,1] vs Fourier MLP results ($p \approx 0.015$) do not survive. That is the honest reading: KAN [1,1] and the Fourier MLP are not distinguishable at matched parameters on this evidence.

5. **What the test does *not* do.** It does not use the within-cell seed spread (the sd over seeds in the Excel file). An alternative would be to test the 45 per-seed differences directly, which gives $t = -10.6$, $p \approx 10^{-13}$ for the worked example, but that treats seeds as independent replicates of the *cell* and inflates the evidence (pseudo-replication: three seeds on the same data are not three new data sets). Averaging over seeds first, as we do, is the conservative choice.

6. **Effect size over p-value.** With $n = 15$ a p-value is a blunt instrument; the quantities to carry into the paper are the mean difference with its interval and d_z. For KAN [1,1] vs MLP of t: -0.020 $[-0.026, -0.013]$, $d_z = -1.7$; for KAN [1,h,1] vs KAN [1,1]: $+0.019$ $[+0.012, +0.025]$, $d_z = 1.6$; for KAN [1,1] vs Fourier MLP: -0.005 $[-0.009, -0.001]$, $d_z = -0.7$.

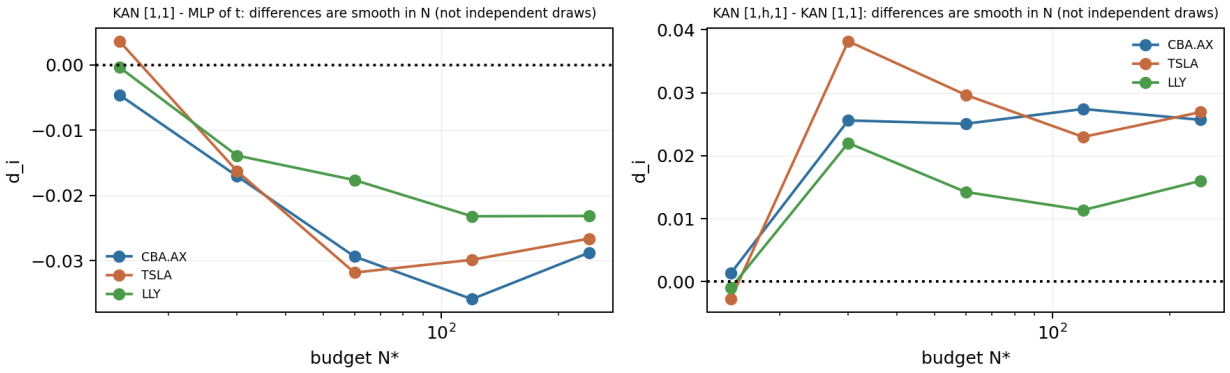


Figure 7: Why independence fails: within each asset the five differences trace a smooth curve in the budget, because they come from the same series at increasing resolution. The t-test treats these points as unrelated draws.

7 One sentence for the paper

"For each comparison a paired t-test was applied to the 15 cell differences (three assets × five budgets, each the mean over three seeds); because the cells share series and are nested in budget, the resulting p-values are reported as descriptive summaries of consistency rather than as inferential

statements, and are accompanied by the mean difference with its 95% interval. A Wilcoxon signed-rank test gave the same conclusions in every case."

References. Student (W. S. Gosset), The probable error of a mean, *Biometrika* 6 (1908) 1–25. R. A. Fisher, Applications of "Student's" distribution, *Metron* 5 (1925) 90–104. W. G. Cochran, The distribution of quadratic forms in a normal system, *Proc. Cambridge Phil. Soc.* 30 (1934) 178–191. F. Wilcoxon, Individual comparisons by ranking methods, *Biometrics Bulletin* 1 (1945) 80–83. J. Cohen, *Statistical Power Analysis for the Behavioral Sciences*, 2nd ed., 1988. S. H. Hurlbert, Pseudoreplication and the design of ecological field experiments, *Ecological Monographs* 54 (1984) 187–211. T. G. Dietterich, Approximate statistical tests for comparing supervised classification learning algorithms, *Neural Computation* 10 (1998) 1895–1923. J. Demšar, Statistical comparisons of classifiers over multiple data sets, *JMLR* 7 (2006) 1–30.